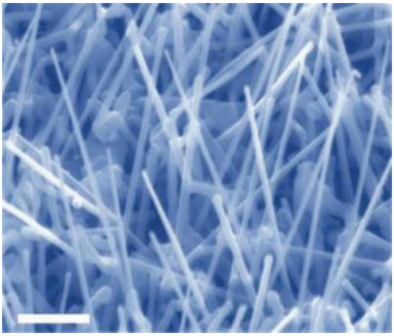
erties of nanomaterials to differ significantly from other materials: increased relative surface area and quantum effects. These factors can change or enhance properties such as reactivity, strength and electrical characteristics, particularly as the structure or particle size approaches the smaller end of the nanoscale.



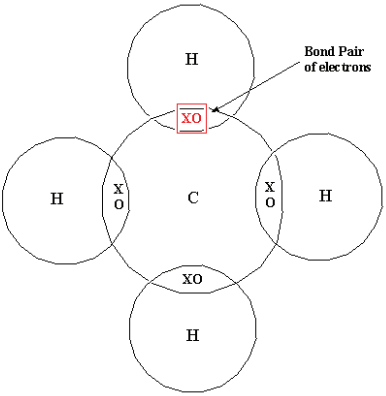
Nanowires – an example of 2d Nanomaterial

To understand how nanomaterials will be used, you need a clear look not only at how they're formed but also at their various configurations. In this section, we look at nano building blocks, and how they're currently being used to enhance all kinds of materials and products. While some of this work is still in the research phase, other work is already being employed in the consumer products and applications industry.

Carbon

Carbon can be regarded as the ubiquitous building block of nanomaterials, and it isn't difficult to see why. Carbon atoms can be found in millions of molecules. These molecules have a wide range of properties, and there are three main reasons behind it:

1. Carbon atoms can bond together with many types of atoms using a process called covalent bonding. In covalent bonding, the atoms that bond (and are held together in a molecule) share two electrons. If the ability of each atom to attract all those negatively charged electrons (called electronegativity) is reasonably close (that is, if the difference in electronegativity is no more than 2), then they can form covalent bonds. Because the electronegativity of carbon atoms is 2.5 (roughly in the mid-range), they can form



Covalent Bonds of Carbon